

Due Date: 18.12.25

Q1. Draw the NPDA of the given below language.

$$L = \{c^n d^m e^{n+m} : n \geq 0, m \geq 0\}$$

$$L = \{a^m b^n c^p d^q \mid m+n=p+q; m, n, p, q \geq 1\}$$

Q2. Convert the grammar into CNF and GNF.

$$A \rightarrow aaAbb \mid bbAaa \mid aaaBBB \mid bbBbb$$

$$B \rightarrow aBaa \mid bbb \mid bbb \mid aaa \mid a$$

Q3. Simplify the grammar.

$$S \rightarrow ASBa \mid BSAb \mid bbb \mid aaa \mid cC$$

$$A \rightarrow BaAB \mid AB \mid a \mid \Lambda$$

$$B \rightarrow BA \mid baabB \mid aabb \mid \Lambda$$

$$C \rightarrow cCc$$

Q4. Convert the Grammar into NPDA by using Stack. Find the string by derivation.

$$S \rightarrow aABB \mid aAA$$

$$A \rightarrow aBB \mid a \mid b$$

$$B \rightarrow bBB \mid A \mid ab$$